Ground News — Complete Platform Documentation
For Product Research & Clone Development

> **Document Purpose:** This document serves as a full-spec reference for every feature, function, page, user flow, data model, and UI element observed on the Ground News platform at `ground.news`. It is intended to be used as a blueprint for building a clone or derivative news aggregation + media bias analysis product.

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1. Product Overview

Ground News is a news aggregation and media bias transparency platform. Its core value proposition is the ability to see how different news outlets — across the political spectrum — are covering the **same story**, allowing users to identify echo chambers, media blind spots, and political framing differences.

Key Stats (as of 2026)
Metric	Value
News Sources Tracked	50,000+
Articles Processed Daily	Tens of thousands
Availability	Web, iOS, Android, Browser Extension
Business Model	Freemium (Free + Paid tiers)
Headquarters	Canada
Founded	2018
Founders	Harleen Kaur, Sukh Singh

What Makes It Unique
- **Not a single news source** — it aggregates from thousands of publishers
- **Does not host full-text articles** — it links to original publishers (respecting copyright, driving traffic)
- **Exposes media bias visually** — color-coded left/center/right spectrum
- **Surfaces "blind spots"** — stories that one side of the spectrum ignores
- **Tracks your own reading bias** — personal news diet analytics
- **Shows media ownership** — reveals who funds/owns which outlets

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2. Company Background

Field	Detail
Company Name	Ground News Inc.
Founded	2018
Founders	Harleen Kaur (CEO) & Sukh Singh (CTO)
Origin Country	Canada
App Package ID (Android)	`com.checkitt`
Apple App Store ID	`1324203419`
Twitter/X Handle	`@Ground_app`
GTM Container	`GTM-WJ94T6S5`
Sentry Environment	`prod`
CDN	BunnyCDN (`groundnews.b-cdn.net`, `grnd.b-cdn.net`)
Tag Manager	Google Tag Manager (custom proxy `gtm.ground.news`)

Company Pages
- `/about` — About the company
- `/history` — Company history
- `/mission` — Mission statement
- `/blog` — Editorial blog

- `/testimonials` — User testimonials
- `/careers` — Job listings
- `/affiliates` — Affiliate program
- `/contact-us` — Contact form

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3. Core Philosophy & Mission

Ground News is built on the conviction that:

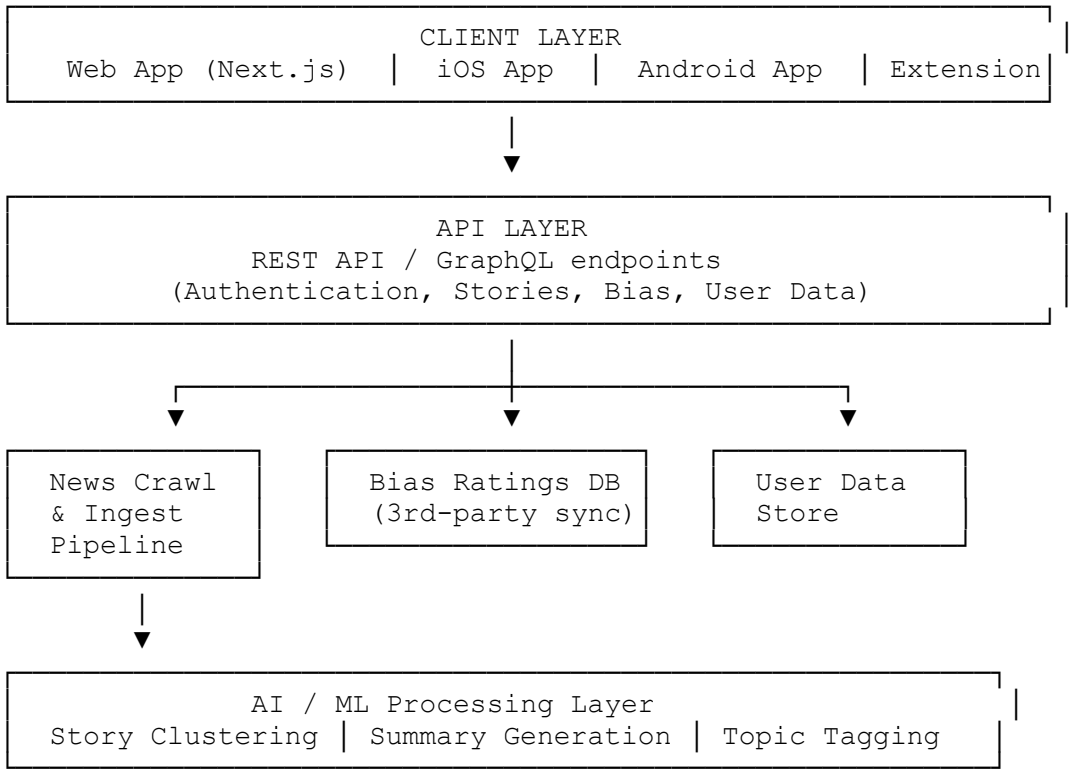
1. **Media bias is real and measurable** — news outlets consistently frame stories based on political leanings
2. **Exposure to one-sided coverage creates blind spots** — readers who only consume left-leaning or right-leaning media miss entire categories of important news
3. **Transparency is the solution** — showing users which outlets are covering what, and how they frame it, empowers critical thinking
4. **No editorial stance** — the platform does not produce its own journalism; it surfaces and contextualizes third-party reporting

Design Principles (Inferred)
- **Show, don't tell** — visual indicators (colors, bars, charts) communicate bias at a glance
- **Link, don't duplicate** — always route users to original publishers
- **Aggregate from neutral third parties** — bias ratings come from external, independent organizations
- **Personal accountability** — help readers see their own consumption patterns
- **Global perspective** — multiple country editions, international coverage

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4. Platform Architecture

High-Level Architecture



Technology Observations (from HTML source)
- **Frontend Framework:** Next.js (React-based SSR/SSG)
- **CSS Framework:** Tailwind CSS (utility classes visible in source)
- **Font:** Universal Sans (custom font, loaded via CSS module)
- **Icons:** Font Awesome (`@fortawesome/fontawesome-svg-core`)
- **Error Monitoring:** Sentry (`sentry-environment=prod`)
- **CDN:** BunnyCDN for images and assets
- **Analytics/Tag Management:** Google Tag Manager (proxied via `gtm.ground.news`)
- **Dark Mode:** System-preference aware + user toggle (`localStorage` key: `theme`)
- **Image Optimization:** Next.js built-in `/_next/image` with responsive srcset
- **Routing:** File-based Next.js routing with named path segments

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5. User-Facing Pages & Routes

Public Routes (No Auth Required)

Route	Description
`/`	Homepage — trending stories feed (default US edition)

`/local`	Local news feed (location-based)
`/blindspot`	Blindspot feed — stories underreported by one political side
`/interest/:slug`	Topic/interest page (e.g., `/interest/ai`, `/interest/water`)
`/subscribe`	Subscription / pricing page
`/login`	User login
`/free-trial-landing`	Free trial offer landing page
`/gift`	Gift a subscription
`/extension`	Browser extension marketing/download page
`/about`	About Ground News
`/history`	Company history
`/mission`	Mission statement
`/blog`	Blog articles
`/testimonials`	User testimonials
`/careers`	Open roles
`/affiliates`	Affiliate program
`/contact-us`	Contact form
`/frequently-asked-questions`	FAQ page
`/rating-system`	Explanation of media bias ratings methodology
`/rating-system#factualityCategory`	Ownership and factuality ratings section
`/group-subscriptions`	Enterprise/group plans

Authenticated Routes

Route	Description
`/my`	"For You" — personalized news feed
`/my-news-bias`	Personal news bias dashboard
`/checkout/referral`	Referral code management

Story Routes (Inferred)
Route	Description
`/interest/:story-slug`	Individual story/event page with full coverage analysis

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6. Core Features — Feed System

6.1 Homepage Feed (Top Stories)

The homepage (`/`) is the primary news feed. It displays stories ranked by editorial importance and trending coverage volume.

Feed Variants:
- **Top Stories** — globally/nationally trending news
- **For You** — personalized feed based on user's interest selections (authenticated users)
- **Local** — geolocation-based local news
- **Blindspot** — politically asymmetric coverage

Story Card — Displayed Information:
Each story card in the feed shows:
- **Headline** — primary news headline (typically neutral/synthesized)
- **Thumbnail image** — associated story image

- **Source Count** — number of outlets covering the story (e.g., "142 sources")
- **Bias Distribution Bar** — horizontal color-coded bar showing Left / Center / Right coverage split
- **Timestamp** — time since story broke
- **Topic Tags** — interest category labels (e.g., "Politics," "Economy")
- **Country/Edition indicator** — which edition surfaced the story

Feed Filtering (Inferred from UI):
- Filter by political leaning (show only Left/Center/Right sources)
- Filter by factuality score
- Filter by geographic region
- Filter by topic/interest
- Filter by time (newest / most discussed)

6.2 Feed Algorithm (Conceptual)

Stories are surfaced based on:
1. **Coverage volume** — number of sources reporting on a topic
2. **Coverage velocity** — how rapidly coverage is growing
3. **Political spectrum diversity** — stories covered across multiple biases get priority
4. **User interest graph** — personalization signals (logged-in users)
5. **Geographic relevance** — edition-based filtering (US, UK, Canada, etc.)
6. **Recency** — time-decay factor on older stories

6.3 Story Clustering

Ground News clusters individual articles from different outlets into a single "story" or "event." This is done via:
- **URL/content deduplication** — preventing the same article from counting multiple times
- **NLP topic matching** — grouping articles about the same real-world event
- **AI-assisted story grouping** — (confirmed from their blog) machine learning to detect related articles across outlets
- **Human editorial oversight** — editors review clustered stories for quality

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7. Core Features — Bias Ratings System

7.1 Overview

Every news source on Ground News is assigned a **political bias rating**. These are not created in-house — they are sourced from reputable, third-party organizations (see Section 29).

7.2 Bias Scale

Label	Color	Description
Left	Blue	Liberal/progressive-leaning outlet
Lean Left	Light Blue	Slightly left-of-center
Center	Purple/Neutral	Balanced, centrist reporting
Lean Right	Light Red/Orange	Slightly right-of-center

| **Right** | Red | Conservative-leaning outlet |

7.3 Bias Distribution Bar

On every story card and story page, a visual bar shows the **percentage of coverage** coming from each political category.

```
[████████▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒]
 Left(45%)  Center(20%)  Right(35%)
```

This bar:
- Is color-coded (blue = left, gray/white = center, red = right)
- Shows percentages on hover/tap
- Updates dynamically as new articles are added to a story
- Gives an instant visual impression of whether a story is being more heavily covered by one side

7.4 Source-Level Bias Display

When viewing a list of articles on a story page:
- Each article row shows the outlet's **bias icon** (colored dot or pill)
- Clicking the outlet name shows the full bias profile for that source
- Users can filter the article list by bias category

7.5 Bias Rating Methodology Page

Route: `/rating-system`

Explains:
- How bias ratings are determined
- Which third-party organizations supply the ratings
- How frequently ratings are updated
- How to suggest a source re-rating

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8. Core Features — Story Page & Coverage Analysis

When a user clicks on any story from a feed, they land on a **Story Page** — the most information-dense view in the product.

8.1 Story Page Sections

Section 1: Story Header
- **Story title** — synthesized neutral headline
- **Thumbnail/hero image**
- **Topic tags** / category breadcrumb
- **Total source count** badge
- **Date/time** of first coverage + last updated
- **Save story** button (bookmarks for later reading)

Section 2: Bias Distribution Panel
- Large, prominent **bias bar** showing Left / Center / Right split
- **Source count** per political category (e.g., "48 Left, 12 Center, 31 Right")
- **Percentage breakdown** as both a visual bar and numeric labels

Section 3: Ground Summary (AI-Generated)
- A **neutral, AI-synthesized summary** of the story
- Pulls key facts from multiple sources across the spectrum
- Designed to be a starting point before reading individual outlet coverage
- Labeled clearly as AI-generated (with human editorial oversight)

Section 4: Bias Comparison Tab
- Side-by-side framing comparison
- Shows **how Left-leaning outlets** describe the event vs. **how Right-leaning outlets** frame it
- Highlights **differences in language, emphasis, and facts included/omitted**
- Educational tool for demonstrating media framing effects

Section 5: Article List / Coverage Feed
- A chronological or filtered list of all articles covering this story
- Each article row includes:
  - **Outlet logo and name**
  - **Bias rating dot/pill** (colored by political lean)
  - **Factuality rating** (if Vantage subscriber)
  - **Article headline**
  - **Publication time**
  - **Paywall indicator** (lock icon if paywalled)
  - **Link out** to original article (opens in new tab)
- **Filter bar** at top: All | Left | Center | Right
- **Sort options:** Newest / Most Relevant / By Outlet

Section 6: Coverage Map (Vantage tier)
- Interactive world map showing **which countries' outlets** are covering the story
- Darker/larger dots = more coverage from that region
- Helps identify if a story has global vs. local reach
- Clicking a region filters the article list to that country's sources

Section 7: Related Stories
- A row or section of algorithmically related stories
- Based on topic overlap, keyword matching, entity recognition

8.2 Story Page URL Structure (Inferred)
```
/interest/[topic-slug]/[story-slug]
```
or
```
/story/[story-id]/[story-slug]
```

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9. Core Features — Blindspot Feed

Route: `/blindspot`

9.1 Concept

The **Blindspot Feed** is one of Ground News's most distinctive and viral features. It identifies stories that have **significant coverage on ONE side of the political spectrum** but are **largely or entirely ignored by the other side**.

9.2 How It Works

Detection Algorithm:
1. Count articles per story across Left / Center / Right sources
2. Compute a **coverage asymmetry score** — how unequal coverage is across the spectrum
3. Flag stories above a threshold as "blindspots"
4. Classify the blindspot direction:
   - **Left Blindspot** — story covered heavily by Right, ignored by Left
   - **Right Blindspot** — story covered heavily by Left, ignored by Right

9.3 UI Display

Each Blindspot card shows:
- Story headline and thumbnail
- **Blindspot badge** — indicates which side is "blinded" (e.g., "Left Blindspot" or "Right Blindspot")
- **Coverage bar** — shows the extreme imbalance (e.g., 90% Right, 2% Left)
- Source count per side
- Timestamp

9.4 User Value

- Forces readers to encounter stories **outside their usual echo chamber**
- Particularly valuable for readers who self-identify as consuming only left or only right media
- Creates a sense of "what am I missing?" — high engagement driver
- Shareable as social proof / conversation starter

9.5 Subscription Access

The Blindspot feed is available to **free users** as a teaser, with full unlimited access gated behind a **paid subscription** (typically Premium or Vantage tier).

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10. Core Features — My News Bias Dashboard

Route: `/my-news-bias` (Vantage tier)

10.1 Concept

The **My News Bias** dashboard is a personal analytics view that tracks your own news consumption patterns. It holds a mirror up to the user, showing them their actual reading behavior — not just what they think they read.

10.2 Dashboard Metrics

Metric	Description
Overall Bias Score	Your personal political lean based on outlets read
Bias Breakdown Chart	Pie/donut chart — % of Left/Center/Right articles read
Blind Spot Score	How many stories you're missing from each side
Reading Diversity Score	A composite score of how balanced your news diet is
Top Sources Read	Your most-visited news outlets with their bias labels
Ownership Concentration	How many of your sources are owned by the same parent company
Factuality Average	Average factuality score of sources you read

10.3 Time-Based Analysis

- View stats for: **Last 7 days / 30 days / 90 days / All time**
- Trend lines showing how your reading habits change over time
- Notifications or nudges if bias score shifts significantly

10.4 Recommendations

Based on the dashboard:
- Ground News suggests specific outlets to add to your reading rotation
- Highlights story types you're underexposed to
- "Read this blindspot story" CTA buttons

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11. Core Features — Coverage Map

Available on story pages (Vantage tier) and potentially as a standalone view.

11.1 What It Shows

- An **interactive world map** (likely built with D3.js or Mapbox)
- **Choropleth or dot overlay** indicating which countries have news outlets reporting on a story
- **Density indicates volume** — countries with many articles have stronger visual indicators
- **Clickable regions** — clicking a country filters the article list to that country's sources

11.2 UI Controls
- Zoom in/out
- Pan/drag
- Country hover tooltip showing: country name, number of articles, top outlets
- Toggle to view by **continent** or **country**

11.3 Use Cases
- Identify if a story is getting international attention or only domestic coverage
- Find **international blindspots** — e.g., a US political story that only US outlets cover

- Discover regional framing differences — European vs. American coverage of the same event

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12. Core Features — Ownership & Factuality Ratings

Route: `/rating-system#factualityCategory` (docs page)

12.1 Ownership Data

For each news outlet on the platform, Ground News displays:

Field	Description
Owner Name	Individual or corporation owning the outlet
Parent Company	Conglomerate/media group (e.g., News Corp, Sinclair, Comcast)
Owner Type	Corporate / Individual / Government / Non-profit
Ownership Concentration	Visual indicator of how many outlets share the same owner

Why it matters:
- Reveals potential editorial conflicts of interest
- Shows when multiple "independent" outlets are actually the same corporate voice
- Exposes government-funded vs. privately funded media

12.2 Factuality Scores

Score Level	Description
Very High	Exemplary factual reporting, strong editorial process
High	Generally accurate with minor errors corrected
Mostly Factual	Generally reliable but with occasional bias or errors
Mixed	Inconsistent factual accuracy
Low	Frequent factual errors or misleading content
Very Low / Conspiracy	Unreliable, conspiracy theories, false information

Factuality data is sourced from **Media Bias Fact Check (MBFC)** and **Ad Fontes Media**.

12.3 Source Profile Card

When a user clicks on any outlet's name, they see a **Source Profile Card** with:
- Outlet name and logo
- Political bias rating + methodology note
- Factuality score
- Ownership information
- Number of articles in Ground News's database from this source
- External links to the outlet's website

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13. Core Features — AI Summaries (Ground Summary)

13.1 Overview

Ground Summary is an AI-generated, multi-source synthesis of a news story. It is available on story pages, generally accessible to all users (free tier).

13.2 How It Works

1. Ground News aggregates all articles covering a story
2. AI model (likely GPT-4 class or similar) reads across multiple articles from different political leanings
3. Generates a **neutral, fact-focused summary** that:
   - Presents core facts agreed upon by most sources
   - Notes **key points of contention** between sources
   - Avoids adopting the framing of any single outlet
4. Human editors review summaries for accuracy and neutrality

13.3 Summary Structure (Inferred)

```
Ground Summary
──────────────────────
[2-3 sentence neutral overview of event]

Key Facts:
• [Fact 1 — agreed upon across sources]
• [Fact 2 — agreed upon across sources]
• [Point of contention — left sources say X, right sources say Y]

What's being debated:
• [Framing difference 1]
• [Framing difference 2]

[AI-generated label + editorial disclaimer]
```

13.4 Subscription Access

- **Free users:** Can view basic Ground Summaries
- **Paid users:** May get longer, more detailed summaries or more frequent AI-powered features

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14. Core Features — Search

14.1 Search Input

Visible in the navigation bar on desktop. Placeholder text:
> *"Enter an article's Web Address, URL, or type to search..."*

This reveals **two distinct search modes:**

Mode 1: URL/Web Address Search
- User pastes a URL from any news article
- Ground News looks up whether it has that article in its database

- If found: shows the story page with bias analysis for that specific article
- If not found: attempts to identify the outlet and provide bias rating
- **Use case:** "I just saw this article on Facebook, is it biased?"

Mode 2: Keyword/Topic Search
- User types a topic, person, event, or news term
- Returns matching stories from the Ground News database
- Results show story cards with bias distribution, source count, etc.

14.2 Search Results Page

- List of matching stories
- Each result shows: headline, thumbnail, bias bar, source count, date
- Filter sidebar: by bias, by date range, by topic
- Sort: Most Relevant / Newest / Most Sources

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15. Core Features — Local News

Route: `/local`

15.1 Overview

The Local News feature surfaces news stories **relevant to the user's geographic area** — city, state/province, or region.

15.2 Location Detection

- **Browser Geolocation API** — prompts user to allow location access
- **Manual city/zip entry** — user types their location
- **IP-based fallback** — rough location estimate without explicit permission
- Location stored in user preferences (authenticated users) or local storage (guests)

15.3 Display

- Same story card format as homepage feed
- Stories filtered to sources in or about the user's local area
- May include **regional/local news outlets** not typically visible in national feeds
- Shows local political context — e.g., local city council coverage, regional events

15.4 "Set Location" Feature

Visible in the top navigation bar (desktop). A button labeled **"Set Location"** opens a dialog to:
- Search for a city/region
- Allow browser geolocation
- Save location preference to account

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16. Core Features — For You / Personalized Feed

Route: `/my` (authenticated users only)

16.1 Overview

The **"For You"** feed is a personalized news stream curated based on:
- **Topic interests** selected during onboarding
- **Reading history** (articles clicked/saved)
- **Source preferences** (sources explicitly followed)
- **Interest pages followed** (e.g., following `/interest/ai`)

16.2 Interest Selection

During onboarding or in settings, users can select topics:
- Politics
- Economy / Business
- Technology / AI
- Science
- Environment / Climate
- Health
- Entertainment
- Sports
- International / World
- ... (dozens more)

16.3 Feed Personalization Algorithm

Combines:
1. **Content-based filtering** — more of what you've clicked
2. **Collaborative filtering** — what similar users read
3. **Diversity injection** — intentionally surfaces articles from underrepresented political leanings in user's history (part of the anti-echo-chamber mission)
4. **Recency boost** — newer stories rank higher
5. **Engagement signals** — save, share, time-on-page

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17. Core Features — Trending Topics & Interests

17.1 Interest Pages

Route: `/interest/:slug`

Examples observed:
- `/interest/water`
- `/interest/ai`
- `/interest/blackrock_0fd268`
- `/interest/floods`
- `/interest/greece`

Each interest/topic page functions like a **topic-specific feed**, showing all stories tagged with that topic.

17.2 Following Topics

Users can **follow** an interest page:
- Followed topics appear in the "For You" feed
- Push notifications can be configured for followed topics

- Following is available free; personalized frequency settings may be premium

17.3 Trending Topics

Displayed in the **footer** (mobile) and potentially as a sidebar widget (desktop):
- Shows currently popular topics/interests on the platform
- Dynamically updated based on article volume and engagement
- Clicking a trending topic goes to its interest page

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18. Browser Extension

Route: `/extension` (marketing page)

18.1 Overview

The **Ground News Browser Extension** (also called the "Bias Checker") brings the platform's analysis directly into the user's browsing experience, without requiring them to visit ground.news.

18.2 Supported Browsers
- Google Chrome
- Mozilla Firefox
- Microsoft Edge
- Opera
- Brave

18.3 Features

Feature	Description
Inline Bias Overlay	When visiting any news article, injects a panel showing the outlet's bias rating and factuality score
Coverage Comparison	Shows how other outlets are covering the same story
Social Media Integration	Works on X/Twitter, Facebook, Reddit, Bluesky, LinkedIn — identifies bias in news links within social feeds
URL Analysis	Paste any article URL to get instant bias context
Save Articles	Bookmark articles to your Ground News account
Citation Generator	Generates formatted citations for research purposes

18.4 Extension Architecture (Inferred)

```
Extension Content Script
    ↓
Detects current page URL
    ↓
Sends URL to Ground News API
    ↓
API returns:
  - Source bias rating
  - Factuality score
  - Related story coverage (list of articles)
```

```
  - Story page link
    ↓
Extension injects UI panel into current page DOM
```

18.5 Permissions Required
- Access to page URLs (to identify article)
- Access to page DOM (to inject UI)
- Network access (to call Ground News API)

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19. Mobile Apps

19.1 iOS App
- **App Store ID:** `1324203419`
- **Store URL:** `https://apps.apple.com/app/apple-store/id1324203419`
- **UTM parameters used:** `pt=107503821`, `ct=unknown`, `mt=8`

19.2 Android App
- **Package ID:** `com.checkitt`
- **Play Store URL:**
`https://play.google.com/store/apps/details?id=com.checkitt`

19.3 App Features

All core web features are mirrored in the apps, plus:
- **Push notifications** — breaking news alerts, blindspot alerts, followed topic updates
- **Offline reading** — save articles for offline access (premium)
- **Haptic feedback** — iOS-specific tactile interactions
- **Widget support** — iOS/Android home screen news widget
- **Dark mode** — follows system preference or manual toggle
- **Biometric login** — Face ID / Touch ID / fingerprint

19.4 App-Exclusive Subscription Plans

Some subscription tiers (particularly lower-cost "Pro" plan) are available primarily through in-app purchase via the App Store / Play Store (subject to platform billing rules).

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20. Subscription & Monetization Model

20.1 Freemium Structure

Ground News uses a **freemium model** — the core experience is free, with premium features gated behind paid tiers.

20.2 Subscription Tiers

Free Tier
- Access to homepage feed, blindspot feed, local news
- Story pages with basic article list
- Bias distribution bar on each story
- Basic AI Ground Summary
- Browser extension (limited)

- Mobile app (limited)
- **No account required** for browsing (account required for personalization)

Pro Tier (~\$9.99/year)
- Everything in Free, plus:
- Enhanced headline comparison across outlets
- Access to more advanced story filtering
- Entry-level paid subscription via mobile app primarily

Premium Tier (~\$29.99/year)
- Everything in Pro, plus:
- **Universal Paywall Filter** — filter out paywalled articles from story lists
- **Hide Read Stories** — removes articles you've already read
- **Burst Your Bubble Newsletter** — exclusive weekly newsletter
- **Unlimited "My Feed" customization**
- **Personal story recommendations**
- Enhanced interest following

Vantage Tier (~\$99.99/year)
- Everything in Premium, plus:
- **My News Bias Dashboard** — full personal reading analytics
- **Ownership Transparency** — who owns each news outlet
- **Factuality Ratings** — full factuality scores for all sources
- **Global Coverage Map** — geographic coverage visualization on story pages
- **Unlimited saved stories** (bookmarks)
- **Full Bias Comparison** feature
- **Advanced analytics** on reading habits

20.3 Group Subscriptions

Route: `/group-subscriptions`

- Designed for: schools, universities, newsrooms, corporations, government agencies
- Volume pricing (per-seat or flat-fee)
- Centralized billing
- Admin dashboard for managing team members
- Custom onboarding materials
- Possibly custom branding/white-label options

20.4 Gift Subscriptions

Route: `/gift`

- Purchase a subscription as a gift for another person
- Select tier and duration
- Recipient receives a redemption link

- No existing Ground News account required to receive a gift

20.5 Free Trial

Route: `/free-trial-landing`

- Limited-time trial of premium features (typically 7–14 days)
- May require payment method on file (auto-renews) or no-card-required trial
- Promoted aggressively via homepage banner and footer

20.6 Revenue Streams Summary

Source	Model
Vantage subscriptions	Annual/monthly recurring
Premium subscriptions	Annual/monthly recurring
Pro subscriptions	Annual/monthly recurring (app stores)
Group subscriptions	Per-seat or flat-fee enterprise
Gift subscriptions	One-time payment
Affiliate commissions	Third-party referrals
Browser extension (indirect)	Drives subscription conversions

---

21. User Account System

21.1 Authentication

- **Sign up** with email + password
- **Login** with email + password
- **Social login** — likely Google / Apple (inferred from mobile app patterns)
- **Biometric** — Face ID / Touch ID (mobile app)

21.2 Account Routes

Route	Description
`/login`	Login page
`/subscribe`	Subscribe / create account + payment
`/checkout/referral`	Apply referral code
`/my`	Authenticated home feed

21.3 User Preferences

Stored per account:
- **Edition preference** — US, UK, Canada, Australia, etc.
- **Theme** — Light / Dark / System
- **Location** — for local news
- **Followed interests/topics**
- **Reading history** (for bias dashboard + recommendations)
- **Notification preferences** — push, email alerts
- **Saved stories** — bookmarks
- **Newsletter subscriptions** — Burst Your Bubble, etc.

21.4 User Onboarding Flow (Inferred)

```
Sign Up → Email Verification → Select Plan (or continue Free)
    → Select Interests (5–10 topics)
    → Set Location (optional)
    → Enable Notifications (optional)
    → Tour of key features (modal walkthrough)
    → Land on "For You" feed
```

---

22. Navigation & UI Structure

22.1 Navigation Bar (Desktop)

Top Utility Bar (dark background, small text):
- Browser Extension link
- Current date/time
- Set Location button
- Edition selector (flag + country name + dropdown)

Main Nav Bar (white/dark bg, larger):
- Ground News logo (left)
- Hamburger menu icon (for sidebar or megamenu)
- Primary nav links:
  - **Home** (active indicator = bold border-bottom)
  - **For You** (authenticated users only)
  - **Local**
  - **Blindspot**
- Search bar (desktop — inline; label wraps hidden input)
- **Subscribe** button (filled, dark)
- **Login** button (outlined)

22.2 Navigation Bar (Mobile)

- Hamburger menu icon (left)
- Back button (appears contextually during navigation)
- Ground News logo (center)
- Subscribe button (right)
- Login button (right)

22.3 Sticky Header Behavior

- **Desktop:** The top utility bar scrolls off; the main nav bar becomes sticky
- **Mobile:** The entire nav bar is sticky, compact
- A sticky promotional banner ("See every side of every news story" + "Get Started" button) appears above the nav bar site-wide for unauthenticated users

22.4 Footer Structure

Column 1 — Company:
About, History, Mission, Blog, Testimonials, Group Subscriptions, Subscribe, Gift, Free Trial, Careers, Affiliates

Column 2 — Help:

Help Center (external: `help.ground.news`), FAQ, Contact Us, Media Bias Ratings, Ownership and Factuality Ratings, Referral Code

Footer Bottom:
- Ground News logo
- App Store badges (Apple App Store, Google Play Store)
- Edition selector
- Social links (Twitter/X `@Ground_app`)
- Copyright notice
- Privacy Policy link
- Terms of Service link

---

23. Editions / Localization System

23.1 Overview

Ground News supports multiple **country editions**, allowing users to see news relevant to their country while still having access to global stories.

23.2 Available Editions (Observed/Inferred)

Edition	Flag	Notes
United States	US	Default for international visitors
United Kingdom	GB	
Canada	CA	Company's home country
Australia	AU	
India	IN	Large English-speaking market
...		Additional editions

23.3 Edition Switching

- Available via **Edition dropdown** in:
  - Desktop top utility bar
  - Footer (mobile and desktop)
- Switching edition updates the feed to show stories prioritized for that country
- Preference saved to account or `localStorage`

23.4 Localization

- Interface language: **English only** (as of 2026)
- News sources: multilingual sources aggregated, but summaries/UI in English
- Currency for subscriptions: USD (may vary by region in app stores)

---

24. Newsletters

24.1 Burst Your Bubble Newsletter

Ground News's flagship email newsletter for premium/Vantage subscribers.

Content:
- Curated selection of stories that represent political blind spots
- "Here's what the left is saying" / "Here's what the right is saying" format
- Designed to burst the user's ideological bubble
- Published weekly (frequency may vary)

Delivery: Email
Access: Premium and Vantage tiers

24.2 Other Newsletters (Inferred)

- Breaking news digests (daily)
- Topic-based newsletters aligned with followed interests
- Promotional/subscription retention emails

---

25. Group Subscriptions

Route: `/group-subscriptions`

Target Audiences
- **Education:** Schools, universities, journalism programs
- **Enterprise:** Corporate communications teams, legal research
- **Government:** Policy research, public affairs offices
- **Media:** Newsrooms, fact-checking organizations
- **Libraries:** Institutional access for patrons

Features (Inferred)
- Admin dashboard to manage seats
- Bulk license purchasing
- Centralized billing / invoicing
- Single Sign-On (SSO) integration (enterprise)
- Usage analytics for admins
- Custom onboarding support

---

26. Referral & Affiliate System

Referral Program

Route: `/checkout/referral`

- Users generate a **personal referral code/link**
- When a friend subscribes using the code, referrer gets a reward (discount, account credit, or free months)
- Managed via the user's account dashboard

Affiliate Program

Route: `/affiliates`

- Third-party publishers, bloggers, or influencers can join
- Earn commission (percentage of subscription value) for every paying subscriber referred
- Tracked via unique affiliate links

- Payouts via standard affiliate network (likely Impact, ShareASale, or custom)

---

27. Gifting System

Route: `/gift`

Flow
1. Select subscription tier (Premium or Vantage)
2. Select duration (1 month / 3 months / 1 year)
3. Enter recipient's email OR get a gift code to share
4. Enter payment details
5. Recipient gets email with redemption instructions
6. Recipient creates account (or links to existing) and activates gift

Gift Card Design
- Downloadable/printable gift card (inferred)
- Shareable digital gift card

---

28. Rating System Documentation

Route: `/rating-system`

This is Ground News's public-facing explanation of how bias and factuality ratings work. It serves both as transparency documentation and as a trust-building marketing asset.

Content Sections

1. **What is Media Bias?** — Definition and why it matters
2. **How Are Bias Ratings Assigned?** — Methodology, third-party sources
3. **The Political Spectrum Explained** — What Left/Center/Right means in the Ground News context
4. **Factuality Ratings** — How reliability is measured
5. **Ownership Data** — How ownership information is gathered and verified
6. **How to Suggest a Re-rating** — Process for contesting or requesting review of a source's rating
7. **Frequently Asked Questions** — About the rating system specifically

---

29. Third-Party Data Partners

Ground News does **not** create its own bias ratings. It aggregates from these respected, independent organizations:

Organization	Data Provided	Website
AllSides	Political bias ratings (community + editorial method)	allsides.com
Ad Fontes Media	Bias + reliability scores (Reliability Chart)	adfontesmedia.com

| **Media Bias Fact Check (MBFC)** | Factuality scores + political bias ratings | mediabiasfactcheck.com |

Data Fusion Strategy
When a source has ratings from multiple organizations:
- Ground News may average/weight the ratings
- May show all three ratings separately and let users see discrepancies
- More data points = higher confidence in the assigned rating

---

30. Data Models (Conceptual)

These are the core data entities the platform requires:

30.1 Source (News Outlet)

```
Source {
  id: string
  name: string
  domain: string           // e.g., "nytimes.com"
  logo_url: string
  bias_rating: enum        // LEFT | LEAN_LEFT | CENTER | LEAN_RIGHT | RIGHT
  bias_score: float        // -10.0 (far left) to +10.0 (far right)
  factuality_score: enum   // VERY_HIGH | HIGH | MOSTLY_FACTUAL | MIXED | LOW | VERY_LOW
  owner_name: string
  parent_company: string
  owner_type: enum         // CORPORATE | INDIVIDUAL | GOVERNMENT | NONPROFIT
  country: string          // ISO 3166-1 alpha-2
  language: string         // ISO 639-1
  is_active: boolean
  created_at: datetime
  updated_at: datetime
}
```

30.2 Article

```
Article {
  id: string
  source_id: string        // FK → Source
  story_id: string         // FK → Story (clustered event)
  url: string              // Original article URL (canonical)
  headline: string
  description: string
  thumbnail_url: string
  published_at: datetime
  is_paywalled: boolean
  word_count: int
  author: string
  created_at: datetime
}
```

30.3 Story (Clustered Event)

```
Story {
  id: string
  slug: string             // URL-friendly identifier
  headline: string         // Synthesized neutral headline
  summary: string          // AI-generated Ground Summary
  thumbnail_url: string
  first_published_at: datetime
  last_updated_at: datetime
  article_count: int
  left_count: int          // # articles from Left sources
  center_count: int
  right_count: int
  left_pct: float
  center_pct: float
  right_pct: float
  is_blindspot: boolean
  blindspot_direction: enum // LEFT_BLINDSPOT | RIGHT_BLINDSPOT | null
  topics: string[]         // FK → Topic[]
  editions: string[]       // Which country editions this appears in
}
```

30.4 Topic / Interest

```
Topic {
  id: string
  slug: string             // e.g., "artificial-intelligence"
  name: string             // e.g., "Artificial Intelligence"
  description: string
  thumbnail_url: string
  follower_count: int
  story_count: int
}
```

30.5 User

```
User {
  id: string
  email: string
  password_hash: string
  display_name: string
  avatar_url: string
  subscription_tier: enum  // FREE | PRO | PREMIUM | VANTAGE
  subscription_expires_at: datetime
  edition: string          // country code
  location: GeoPoint
  theme: enum              // LIGHT | DARK | SYSTEM
  followed_topics: string[] // FK → Topic[]
  created_at: datetime
  last_login_at: datetime
}
```

```

### 30.6 UserReadingHistory

```
UserReadingHistory {
  id: string
  user_id: string   // FK → User
  article_id: string   // FK → Article
  story_id: string   // FK → Story
  source_id: string   // FK → Source
  read_at: datetime
  time_on_page_seconds: int
  clicked_out: boolean   // Did user click through to original article?
}
```

### 30.7 SavedStory

```
SavedStory {
  id: string
  user_id: string
  story_id: string
  saved_at: datetime
  notes: string            // User notes (if any)
}
```

---

31. API Layer (Inferred)

Based on observed behavior and standard news aggregation platforms:

31.1 Core Endpoints

```
GET  /api/feed/top?edition={country}&page={n}&limit={n}
GET  /api/feed/for-you?userId={id}&page={n}&limit={n}
GET  /api/feed/local?lat={f}&lon={f}&radius={km}
GET  /api/feed/blindspot?direction={left|right|all}&page={n}
GET  /api/stories/{storyId}
GET  /api/stories/{storyId}/articles?bias={l|c|r}&sort={new|rel}
GET  /api/stories/{storyId}/summary
GET  /api/stories/{storyId}/coverage-map
GET  /api/stories/search?q={query}&edition={country}
GET  /api/sources/{sourceId}
GET  /api/sources/search?q={query}
GET  /api/topics
GET  /api/topics/{topicSlug}/stories?page={n}
GET  /api/user/bias-dashboard
GET  /api/user/saved-stories
GET  /api/user/preferences
PUT  /api/user/preferences
POST /api/user/saved-stories
DELETE /api/user/saved-stories/{storyId}
POST /api/auth/login
```
```

```
POST /api/auth/signup
POST /api/auth/logout
POST /api/auth/refresh
```

31.2 Browser Extension API

```
GET  /api/extension/analyze?url={encoded-article-url}
  → Returns: source bias, story context, related articles
```

---

32. Tech Stack (Observed & Inferred)

Frontend
Layer	Technology
Framework	Next.js 14+ (React, SSR + ISR)
Styling	Tailwind CSS
UI Font	Universal Sans (custom)
Icons	Font Awesome (SVG inline)
Map	Likely D3.js, Mapbox, or Leaflet (coverage map)
Charts	Likely Recharts, Chart.js, or D3 (bias bars, dashboards)

Backend (Inferred)
Layer	Technology
API	Node.js / Next.js API routes or separate service
Database	PostgreSQL (primary relational data)
Cache	Redis (feed caching, session storage)
Search	Elasticsearch or Algolia (article/story search)
Queue	Bull/Redis or SQS (article ingestion pipeline)
AI/ML	OpenAI API or fine-tuned model (story summaries)
Storage	S3 or equivalent (images, thumbnails)

Infrastructure
Layer	Technology
CDN	BunnyCDN (`groundnews.b-cdn.net`)
Hosting	Vercel (Next.js), or AWS/GCP
Tag Management	Google Tag Manager (proxied)
Error Monitoring	Sentry
Analytics	Google Analytics (via GTM)
Mobile	React Native (likely, given web-first architecture)
App Store Payments	Apple In-App Purchase + Google Play Billing

---

33. Design System & UI Tokens

Based on Tailwind class analysis from the source HTML:

33.1 Color Tokens

```css
/* Light Mode */
```

```
--color-light-primary: #EEEFE9  /* Off-white — page background */
--color-dark-primary:  #262626  /* Near-black — text, header bg */
--color-light-heavy:   /* Medium gray — borders, muted text */
--color-focus:         /* Highlight yellow/orange — sticky banner bg */
--gray-200:            /* Placeholder text color */

/* Dark Mode (auto-detected) */
--color-dark-primary:  #EEEFE9  /* Inverted */
--color-light-primary: #262626  /* Inverted */
```

33.2 Political Bias Colors

```css
--color-left:      #3B82F6  /* Blue */
--color-lean-left: #93C5FD  /* Light Blue */
--color-center:    #9CA3AF  /* Gray / Purple-gray */
--color-lean-right:#FCA5A5  /* Light Red */
--color-right:     #EF4444  /* Red */
```

33.3 Spacing Scale (Tailwind custom)

```
px-marginmobile   → Mobile horizontal padding
px-margintablet   → Tablet horizontal padding
px-margindesktop  → Desktop horizontal padding
max-w-screen-designmax → Max container width
```

33.4 Breakpoints

```
mobile:   default (< 768px)
tablet:   md: (768px+)
desktop:  lg: / xl: (1024px+ or 1280px+)
```

33.5 Z-Index Layers

```
z-999   → Navigation bar
z-1000  → Sticky promo banner (above nav)
z-20    → Footer
```

33.6 Typography

```
text-12, text-14, text-15, text-16, text-18, text-20, text-22
→ Custom font-size scale (in rem equivalents)

font-medium, font-bold, font-semibold → Weight variants
font-universal-sans → Primary typeface
```

33.7 Component Classes

```css
/* Button — Primary (Subscribe) */
.btn-primary: bg-dark-primary text-light-primary px-[0.6rem] py-[0.6rem]
              rounded-[4px] font-bold

/* Button — Secondary (Login) */
.btn-secondary: border border-dark-primary text-dark-primary
                px-[0.7rem] py-[0.7rem] rounded-[4px] font-bold

/* Rounded Corners */
rounded-lg-s → Custom small-large radius
rounded-[4px] → Standard button radius
```

---

34. SEO & Metadata

34.1 Meta Tags (Homepage)

```html
<title>Ground News</title>
<meta name="description"
  content="Top Stories from around you and around the world.
           Compare how different news publishers frame the same news"/>

<!-- Open Graph -->
<meta property="og:title" content="Ground News"/>
<meta property="og:description"
  content="Top Stories from around you and around the world.
           Compare how different news publishers frame the same news"/>
<meta property="og:site_name" content="Ground News"/>
<meta property="og:image"
  content="https://groundnews.b-cdn.net/assets/logo4/GN_Square.png"/>

<!-- Twitter Card -->
<meta name="twitter:card" content="summary_large_image"/>
<meta name="twitter:site" content="@Ground_app"/>
<meta name="twitter:title" content="Ground News"/>
<meta name="twitter:description"
  content="The biggest source for breaking news around the world.
           Compare headlines across the political spectrum
           using media bias ratings driven by data."/>
<meta name="twitter:image"
  content="https://groundnews.b-cdn.net/assets/web/images/Twitter-Meta-
2.png"/>

<!-- Robots -->
<meta name="robots" content="noimageindex"/>
```

34.2 Story Page SEO (Inferred)

Each story page would have:
- Dynamic `<title>` = Story headline
- Dynamic `og:description` = Ground Summary first sentence
- Dynamic `og:image` = Story thumbnail
- `article:published_time` OpenGraph tag

- Structured data (JSON-LD) for `NewsArticle` schema type

34.3 Favicon

Custom SVG favicon with Ground News "G" logo mark, adapts to light/dark mode via CSS `@media (prefers-color-scheme: dark)`.

---

35. Analytics & Tracking

35.1 Google Tag Manager

- Container ID: `GTM-WJ94T6S5`
- Proxied through `gtm.ground.news` (for privacy/ad-blocker resilience)
- Fires: page views, CTA clicks, subscription events, scroll depth

35.2 Observed Element IDs (for GTM tracking)

Key interactive elements tagged with `data-element-id` or `id` attributes for click tracking:

```
sticky-banner-get-started-button  → Top sticky CTA
header-logo                       → Logo click
header-navbar-home                → Home nav
header-navbar-my                  → For You nav
header-navbar-local               → Local nav
header-navbar-blindspot           → Blindspot nav
header-search                     → Search icon/field
header-subscribe                  → Subscribe button (mobile)
header-login                      → Login button
subscribe-top                     → Subscribe button (desktop)
location-top                      → Set location button
top-edition-clicked               → Edition dropdown
hamburger-mobile-click            → Mobile menu open
header-back-click-mobile          → Mobile back button
browser-extension-top             → Browser extension link (utility bar)
footer_About, footer_History...   → Footer link clicks
footer_Help-Center, footer_FAQ... → Footer help link clicks
apple-app-store                   → App Store badge click
google-play-store                 → Play Store badge click
footer-edition-clicked            → Footer edition switcher
```

35.3 Sentry Error Monitoring

- Environment: `prod`
- Release tracked by git SHA: `bf8449ed5853e81cfd315fc0cc019cd3b438d8d2`
- DSN public key: `a7d1977d3cc340febc6ad1652912d39a`
- Organization ID: `392313`
- Sentry trace headers injected into page metadata

---

36. Accessibility Features

36.1 Observed Accessibility Implementations

- **Skip link:** `<a class="sr-only" href="#main">Skip to main content</a>` — keyboard navigation
- **`aria-label`** on logo link: `"Ground News Home"`
- **`aria-current="page"`** on active nav links
- **`aria-expanded`, `aria-haspopup`, `aria-controls`** on dropdown triggers
- **`aria-hidden="true"`** on decorative SVG icons
- **`role="img"`** with accessible description on functional SVGs
- **`<nav aria-label="Primary">`** — landmark navigation labeling
- **`<time dateTime="...">`** — semantic time elements for screen readers
- **`<section aria-label="Site utilities">`** — landmark sectioning
- **`id="main"`** — main content landmark for skip link target

36.2 Keyboard Navigation

- Tab order follows DOM order (logical reading sequence)
- Focus rings visible on interactive elements (`focus-visible:ring-2`)
- Modals trap focus (location dialog, edition selector)

36.3 Color & Contrast

- Political bias colors (blue/red) are supplemented by labels (not solely color-dependent)
- Light/dark mode respects system preference
- Font size minimum 12px (text-12)

---

37. Footer & Company Links

Complete Footer Link Inventory

Company Section:
- `/about` — About
- `/history` — History
- `/mission` — Mission
- `/blog` — Blog
- `/testimonials` — Testimonials
- `/group-subscriptions` — Group Subscriptions
- `/subscribe` — Subscribe
- `/gift` — Gift
- `/free-trial-landing` — Free Trial
- `/careers` — Careers
- `/affiliates` — Affiliates

Help Section:
- `https://help.ground.news` — Help Center (external)
- `/frequently-asked-questions` — FAQ
- `/contact-us` — Contact Us
- `/rating-system` — Media Bias Ratings
- `/rating-system#factualityCategory` — Ownership and Factuality Ratings
- `/checkout/referral` — Referral Code

App Downloads:
- iOS: `https://apps.apple.com/app/apple-store/id1324203419`
- Android: `https://play.google.com/store/apps/details?id=com.checkitt`

---

38. Clone Development Roadmap

This section outlines a practical phased roadmap for building a Ground News clone from scratch.

Phase 1 — MVP Core (Weeks 1–8)

Goal: A functional news aggregation feed with basic bias indicators.

- [ ] Set up Next.js project with Tailwind CSS
- [ ] Design and implement core data models (Source, Article, Story)
- [ ] Build a news crawler/ingestion pipeline (RSS feeds + scraping)
- [ ] Implement story clustering algorithm (NLP deduplication)
- [ ] Seed database with ~500 sources and their bias ratings (from AllSides, MBFC)
- [ ] Build homepage feed UI (story cards with bias distribution bars)
- [ ] Build story page UI (article list, bias breakdown panel)
- [ ] Implement basic search (keyword)
- [ ] Implement user auth (email/password, JWT)
- [ ] Basic subscription checkout (Stripe integration)
- [ ] Deploy to Vercel + configure CDN

Phase 2 — Core Features (Weeks 9–16)

Goal: Blindspot feed, personalization, and browser extension.

- [ ] Implement Blindspot detection algorithm
- [ ] Build Blindspot feed page
- [ ] Build "For You" personalized feed (interest selection + filtering)
- [ ] Build Local News feed (geolocation integration)
- [ ] Build Interest/Topic pages
- [ ] Implement "Save Story" bookmarking
- [ ] Build browser extension (Chrome, Firefox)
- [ ] Implement URL-based search in extension + web
- [ ] Build mobile apps (React Native)
- [ ] Add push notifications

Phase 3 — Premium Features (Weeks 17–24)

Goal: Full Vantage-equivalent feature set.

- [ ] Build AI Ground Summary (OpenAI API integration)
- [ ] Build My News Bias Dashboard (reading history analytics)
- [ ] Build Ownership data layer + source profile cards
- [ ] Build Factuality Ratings display
- [ ] Build Coverage Map (Mapbox or Leaflet)
- [ ] Implement paywall filter feature
- [ ] Implement "Hide Read" feature
- [ ] Build newsletter system (email templates + scheduling)
- [ ] Build edition/localization system (multi-country)
- [ ] Build group subscription admin dashboard

Phase 4 — Growth & Monetization (Weeks 25–32)

Goal: Full monetization and growth loops.

- [ ] Implement referral system
- [ ] Implement affiliate program
- [ ] Build gift subscription flow
- [ ] SEO optimization (structured data, sitemap, etc.)
- [ ] Performance optimization (ISR, caching, image optimization)
- [ ] A/B testing framework
- [ ] Accessibility audit and fixes
- [ ] Internationalization (i18n framework)
- [ ] Mobile app store submissions

Key Technical Challenges for Clone Development

Challenge	Complexity	Notes
Story Clustering	High	NLP deduplication across 50K+ sources at scale is non-trivial
Crawler at Scale	High	Crawling RSS/HTML from 50K sources requires distributed infrastructure
Bias Rating Data	Medium	Must obtain data from AllSides/MBFC APIs or license agreements
AI Summaries	Medium	OpenAI API is straightforward; maintaining neutrality is the hard part
Real-time Feed Updates	Medium	WebSockets or polling for live story count updates
Coverage Map	Medium	Requires article-level geolocation tagging
My News Bias Dashboard	Medium	Requires reading history tracking + bias aggregation math
Browser Extension	Medium	Cross-browser compatibility + DOM injection complexity
Legal/Copyright	High	Must NOT host article text; headline + summary only
Scaling	High	Feed caching, CDN, database read replicas needed early

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Document compiled from: live site analysis of `ground.news`, web research, Wikipedia, Apple App Store listing, Google Play Store listing, and publicly observable HTML source. Last updated: September 2026.